

L^2 -Norm Two-Sample Mean Tests Under Compound Symmetry: Power Saturation, Decorrelation, and Signal Alignment

Ram Goenka

Department of Statistics, University of Pittsburgh



The problem, and a puzzle

In high-dimension, low-sample-size (HDLSS) data we test $H_0 : \boldsymbol{\mu}_1 = \boldsymbol{\mu}_2$ against $H_1 : \boldsymbol{\mu}_1 \neq \boldsymbol{\mu}_2$ with $p \gg n$, where Hotelling's T^2 is undefined because the pooled covariance is singular. Three L^2 -norm tests avoid inversion by building on $\|\bar{\mathbf{X}}_1 - \bar{\mathbf{X}}_2\|^2$: **Bai–Saranadasa (BS)**, **Chen–Qin (CQ)** and **Srivastava–Du (SD)**.

An initial $4 \times 4 \times 3 \times 4$ grid study ($B = 1000$) found BS and CQ interchangeable and dominant under identity and AR(1) covariance. However, *all three looked powerless under compound symmetry (CS) at $\rho = 0.5$.*

BS power, sparse signal, $n = 20$	$p = 200$	500	2000	10000
Identity	.308	.609	.984	1.000
Compound symmetry ($\rho = 0.5$)	.084	.081	.089	.106
its own null rate	.080	.066	.080	.085

Is that a failure of the tests, or a property of the covariance?

The governing quantity is $\text{tr}(\boldsymbol{\Sigma}^2)$

For the BS/CQ family with equal group sizes $n_1 = n_2 = n$, the local asymptotic power is

$$\text{power} \approx \Phi(-z_\alpha + \text{SNR}), \quad \text{SNR} = \frac{n \|\boldsymbol{\delta}\|^2}{2\sqrt{2} \sqrt{\text{tr} \boldsymbol{\Sigma}^2}}, \quad \boldsymbol{\delta} = \boldsymbol{\mu}_1 - \boldsymbol{\mu}_2.$$

Covariance enters *only* through $\text{tr} \boldsymbol{\Sigma}^2$. Under the equal-norm calibration $\|\boldsymbol{\delta}\|^2 = 0.0125 p$, with $\lambda_1 = 1 + (p-1)\rho$:

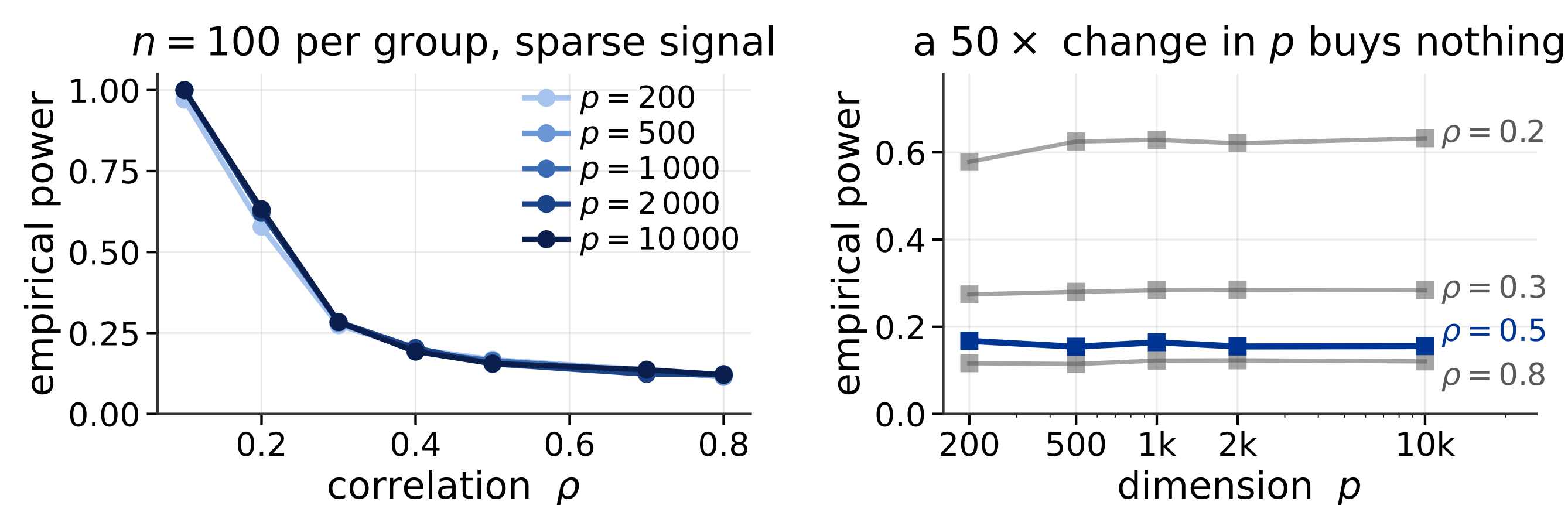
$\boldsymbol{\Sigma}$	$\text{tr} \boldsymbol{\Sigma}^2$	$\text{SNR} \propto$
Identity	p	$n\sqrt{p}$
AR(1), $\rho = 0.5$	$\frac{1+\rho^2}{1-\rho^2} p = \frac{5}{3}p$	$0.77 n\sqrt{p}$
CS, ρ	$\lambda_1^2 + (p-1)(1-\rho)^2 \approx \rho^2 p^2$	n

Under CS the standard deviation of the statistic grows like p rather than $\sqrt{p}$, exactly cancelling the linear growth of the signal:

$$\text{SNR}_{\text{CS}} = \frac{n \cdot 0.0125 p}{2\sqrt{2} \rho p} = \frac{0.0125 n}{2\sqrt{2} \rho} \quad \text{free of } p \text{ for fixed } \rho > 0.$$

Not powerless, but saturated. Only the dimension-driven gain is lost. Solving $\text{SNR} = z_{0.05}$ gives $n \approx 186$ for 50% power at $\rho = 0.5$, far outside the original grid.

The dimension-driven gain does saturate



Raw BS power under CS, $B = 5000$ per cell. Left: the five dimensions are indistinguishable. Right: power against p on a log scale at fixed ρ .

At $n = 100$ the curves for $p = 500$ and $p = 10000$ are nearly identical (0.625 vs. 0.632 at $\rho = 0.2$; 0.154 vs. 0.156 at $\rho = 0.5$). Under identity covariance the same design gives $0.168 \rightarrow 0.992$ over that range of p .

The null is not Gaussian either

Chen–Qin require $\text{tr}(\boldsymbol{\Sigma}^4) = o(\text{tr}^2(\boldsymbol{\Sigma}^2))$. Under fixed- ρ compound symmetry that condition fails by the widest possible margin:

$$\frac{\text{tr}(\boldsymbol{\Sigma}^4)}{(\text{tr}(\boldsymbol{\Sigma}^2))^2} \rightarrow \frac{(\rho p)^4}{((\rho p)^2)^2} = 1, \quad \text{equivalently} \quad \frac{\lambda_1^2}{\text{tr}(\boldsymbol{\Sigma}^2)} \rightarrow 1.$$

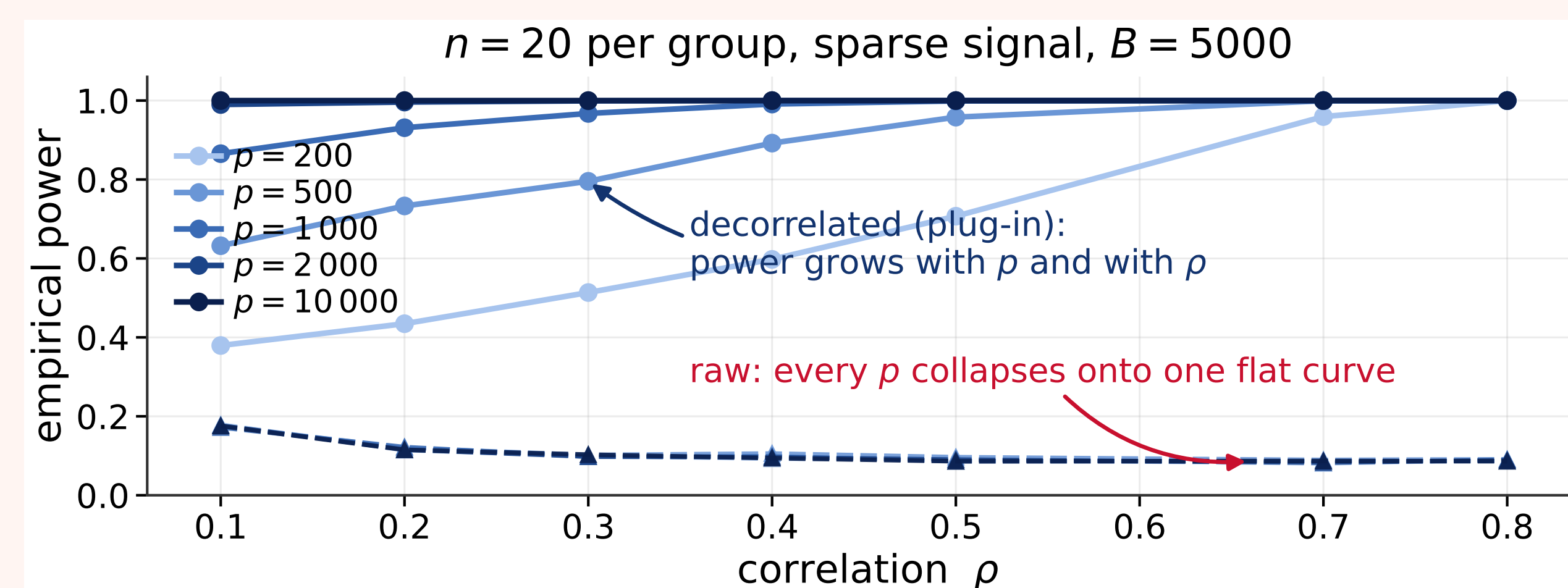
Fixed- ρ CS is an **extreme strongly spiked eigenvalue (SSE) model**, outside the regime that yields Gaussian limits here. The standardized null approaches $(\chi_1^2 - 1)/\sqrt{2}$, and referring that rule to the normal 5% critical value gives the benchmark $\Pr(\chi_1^2 > 1 + 1.645\sqrt{2}) = 0.0682$.

Decorrelation: a closed-form $O(np)$ fix

CS has a known rank-one eigensystem, so the whitening map is explicit:

$$\boldsymbol{\Sigma}^{-1/2} = a\mathbf{P}_\perp + b\mathbf{P}_1, \quad a = (1-\rho)^{-1/2}, \quad b = \lambda_1^{-1/2},$$

applied per row as $\mathbf{x} \mapsto a(\mathbf{x} - \bar{x}\mathbf{1}) + b\bar{x}\mathbf{1}$ in $O(np)$ as *no $p \times p$ object is required*. The plug-in uses the mean off-diagonal sample correlation, read off $\|\mathbf{Z}\mathbf{1}_p\|^2/\text{df}$.

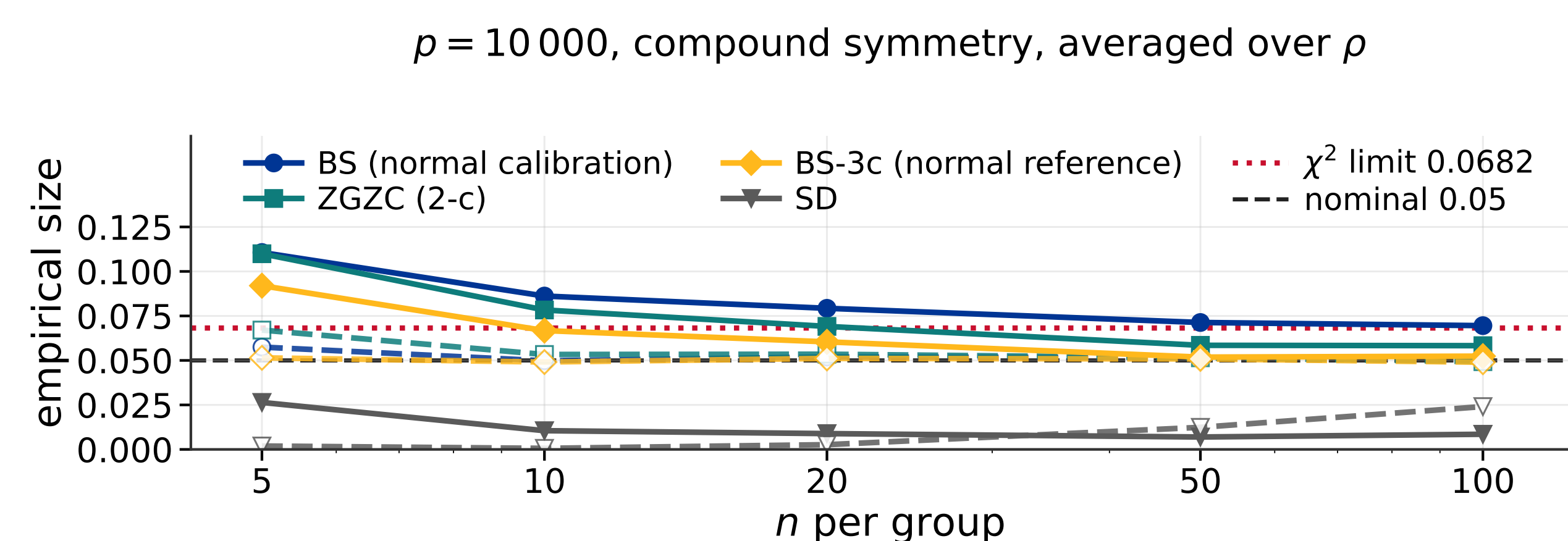


Raw (dashed) and plug-in decorrelated (solid) BS power. Whitening restores an $O(n\sqrt{p})$ SNR through the orthogonal component of the signal.

At $p = 10000$, $n = 5$ power rises from 0.122 to 0.898 and by $n = 20$ it reaches 1.000 at every simulated ρ . Across 210 decorrelated alternative cells, predicted power matches simulation to a mean absolute error of 0.0032 ($r = 0.9998$).

The plug-in is as good as the oracle. For large p the standardized coordinates share the same realized common-factor variance, giving $\text{SD}(\hat{\rho}) \approx \rho(1-\rho)\sqrt{2/(2n-2)}$, matched by simulation to a mean ratio of 1.019 over all 175 (p, n, ρ) cells. A common scalar cancels from the standardized statistic, so only the relative shrinkage $b/a = \sqrt{(1-\hat{\rho})/\hat{\lambda}_1}$ matters, and recovery is robust to error in $\hat{\rho}$ (plug-in 0.898 vs. oracle 0.902 at $p = 10000$, $n = 5$).

Two distinct problems: calibration and SNR



Empirical size averaged over $\rho \in \{0.1, \dots, 0.8\}$; $B = 5000$ per cell (null Monte Carlo SE ≈ 0.003). Solid/filled: raw. Dashed/open: after plug-in decorrelation.

BS and CQ converge to 0.0682, not 0.05. Normal-reference calibrations fix the size. Whitening recovers the SNR, since recalibration changes the critical value but not $\|\boldsymbol{\delta}\|^2/\text{tr} \boldsymbol{\Sigma}^2$. SD stays conservative throughout a separate, p/n -driven difficulty.

Signal alignment, the hidden variable

Decompose the mean difference along the common-factor direction $\mathbf{u} = \mathbf{1}_p/\sqrt{p}$:

$$\boldsymbol{\delta} = c_1 \mathbf{u} + \boldsymbol{\delta}_\perp, \quad \gamma = \frac{c_1^2}{\|\boldsymbol{\delta}\|^2} \in [0, 1].$$

For a shift of equal size on a support fraction s , $\gamma = s$ **exactly**, so the original sparse / moderate / dense designs (5%, 20%, 100%) confounded *sparsity* with *common-factor alignment*. After oracle whitening the transformed signal energy is

$$\|\tilde{\boldsymbol{\delta}}\|^2 = \|\boldsymbol{\delta}\|^2 \left[\frac{\gamma}{\lambda_1} + \frac{1-\gamma}{1-\rho} \right],$$

so whitening helps through $\boldsymbol{\delta}_\perp$ and hurts only as $\gamma \rightarrow 1$. Keeping the spike explicit and treating the bulk as normal gives a finite- n rule,

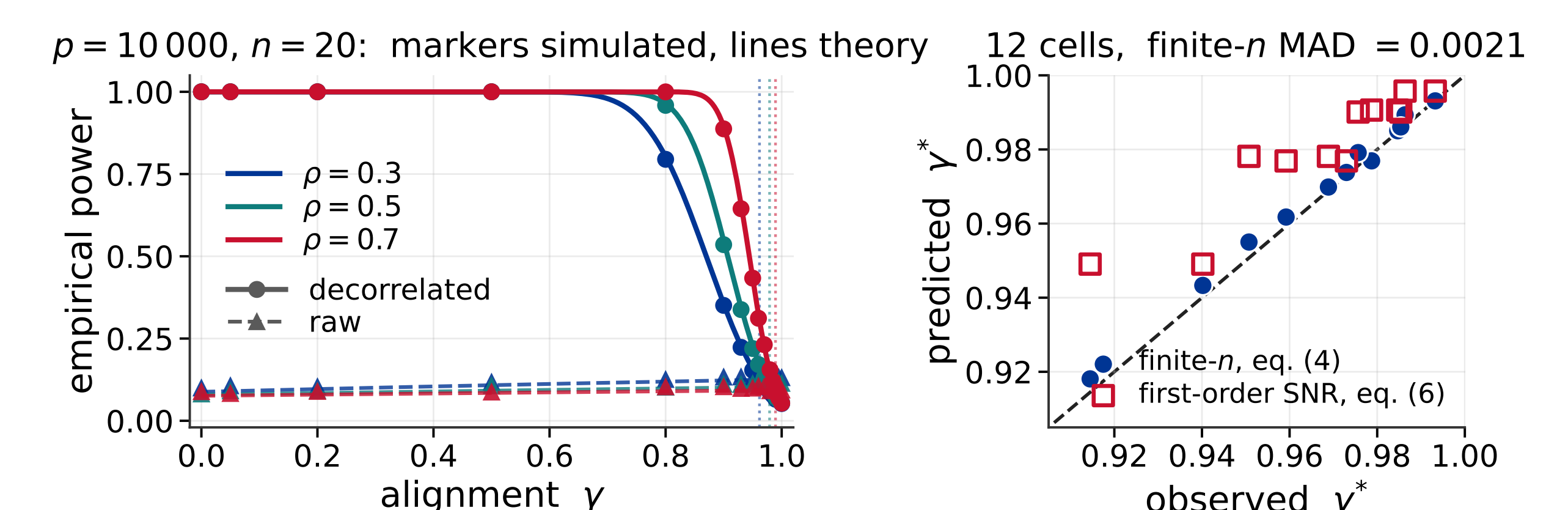
$$M \stackrel{d}{\approx} a_u [(Z+m)^2 - 1] + (1-\gamma)E + s_b V, \quad Z, V \stackrel{\text{indep.}}{\sim} N(0, 1),$$

with $E = \|\boldsymbol{\delta}\|^2$, $a_u = 2\lambda_1/n$, $a_b = 2(1-\rho)/n$, $m^2 = \gamma E/a_u$ and $s_b^2 = 2(p-1)a_b^2 + 4a_b(1-\gamma)E$.

The crossover

Equating the first-order raw and whitened SNRs gives a finite- p threshold in closed form:

$$\gamma_{\text{SNR}}^*(p, \rho) = \frac{(1-\rho)^{-1} - [1 + (p-1)\rho^2]^{-1/2}}{(1-\rho)^{-1} - [1 + (p-1)\rho]^{-1}} = 1 - \frac{1-\rho}{\rho\sqrt{p}} + o(p^{-1/2}).$$



Left: power against alignment γ ; dotted rules mark the predicted crossings. Right: predicted versus observed γ^* over $p \in \{2000, 10000\}$, $n \in \{20, 50\}$, $\rho \in \{0.3, 0.5, 0.7\}$.

The first-order threshold sits above the observed crossings, with the gap shrinking in n .

Take-aways

- 1. Saturation, not failure.** Fixed positive CS cancels the dimension-driven power gain exactly. SNR then depends on n and ρ only.
- 2. Calibration is not power.** Normal-reference tests fix the size, whitening is what recovers the SNR. Here it restores nominal size too.
- 3. The trade-off is heavily one-sided.** Maximum gain from decorrelation 0.858-0.921; maximum loss only 0.040-0.171. Since $\gamma^* \rightarrow 1$, the harmful region sits where the raw test is already weak.

Key references

- 1. Bai, Z. & Saranadasa, H. (1996).** Effect of high dimension: by an example of a two sample problem. *Statistica Sinica* **6**, 311–329.
- 2. Chen, S. X. & Qin, Y.-L. (2010).** A two-sample test for high-dimensional data with applications to gene-set testing. *Ann. Statist.* **38**, 808–835.
- 3. Aoshima, M. & Yata, K. (2018).** Two-sample tests for high-dimension, strongly spiked eigenvalue models. *Statistica Sinica* **28**, 43–62.